

UNG FACILITIES PLANNING & GIS

Version 1.0

May 2026

Inspects why AD records were excluded from the lookup table — no ArcGIS required

AD_Lookup_Diagnostic_V1.0.py

Purpose

A standalone diagnostic tool with no ArcGIS dependency. Runs the same crosswalk and filter logic as the updater's `prepare_lookup_table` function and writes two output files: classified excluded records and a console summary showing which building names are failing the crosswalk. Run between pipeline runs to investigate the excluded record population without touching GIS data.

When to Run

- On demand — not part of the weekly pipeline
- After any run where you want to investigate what is in the 263 excluded records
- When the AD_Lookup output shows a large `no_gis_bldg` count worth investigating
- When new building names appear in AD and you need to identify which ones need crosswalk entries

Requirements

- Python with pandas only — no ArcGIS required
- Parsed AD Excel, `Building_Crosswalk.xlsx`, `Department_to_Domain_CrosswalkV2.xlsx`

Inputs

Item	Value	Notes
Parsed AD Excel	<code>FacStaffWithAddress_MMDDYY YY_parsed.xlsx</code>	Update <code>PARSED_AD_FILE</code> path for each run
Building Crosswalk	<code>Building_Crosswalk.xlsx</code>	Static — rarely changes
Department Crosswalk	<code>Department_to_Domain_CrosswalkV2.xlsx</code>	Static — rarely changes

Excluded Record Classifications

no_gis_bldg: AD building name not found in `Building_Crosswalk.xlsx` — either a typo/nickname/abbreviation, or a building not yet mapped. These are the actionable crosswalk gaps.

no_room: Room field blank or unparseable — employee location was entered incompletely in AD. AD data quality issue.

no_surname: Surname blank in AD — service accounts or data gaps. Rare.

Outputs

Item	Value	Notes
Excluded_AD_Records_TIMES TAMP.xlsx	ActiveDirectoryParsing\	All excluded records with Reason column, sorted by Reason then Building
Console summary	Printed to screen	Count by reason category and top unmatched building names ranked by frequency

Using the Output

Filter the Excluded file to Reason = no_gis_bldg and look at the Building column. Each unique building name that appears frequently is a candidate for a new crosswalk entry. Check whether the building exists in GIS — if it does, add it to Building_Crosswalk.xlsx. If it does not exist in GIS, flag it as an AD data quality issue for the relevant department.